

- C. Necrosis occurs in healthy cells, while apoptosis occurs only in diseased cells.
D. Necrosis always occurs as part of normal physiological processes.
- 70) Which of the following is a reversible cellular injury?
A. Apoptosis
B. Cellular swelling due to hypoxia
C. Necrosis
D. Mitochondrial rupture
- 71) Which of the following is an example of labile cells?
A. Neurons
B. Bone marrow cells
C. Liver hepatocytes
D. Skeletal muscle cells
- 72) Which type of cells has limited regenerative capacity and typically divides only when stimulated?
A. Labile cells
B. Stable cells
C. Permanent cells
D. Transitional cells
- 73) Which of the following cells are considered permanent cells with no significant capacity to regenerate?
A. Columnar epithelium of the intestinal tract
B. Alveolar cells of the lung
C. Skeletal and cardiac muscle cells
D. Transitional epithelium of the urinary tract
- 74) What type of cells are present in the squamous epithelium of the skin, mouth, vagina, and cervix?
A. Stable cells
B. Permanent cells
C. Labile cells
D. Transitional cells
- 75) Which of the following is not an example of stable cells?
A. Epithelium of kidney tubules
B. Liver hepatocytes
C. Alveolar cells of the lung
D. Neurons
- 76) Which of the following is a common application of cytopathology in cancer prevention?
A. Routine blood testing for genetic mutations
B. Examination of tissue sections for structural abnormalities
C. Screening for early detection of asymptomatic cancers, such as cervical cancer through cervical scrapings
D. Radiographic examination of bones for hidden tumors
- 77) Apoptosis has the following features **except**:
A. There is cell shrinkage in apoptosis
B. There are no acute inflammatory cells surrounding apoptosis
C. There may be single cell loss or affect clusters of cells

- D. Apoptosis is seen in pathologic processes only
- 78) In atrophy, the cells are:
- A. Dead cells
 - B. Shrunken cells
 - C. Irreversibly injured cells
 - D. Reversibly injured cells
- 79) For metaplasia the following holds true:
- A. It is a disordered growth
 - B. It affects only epithelial tissues
 - C. It is a reversible change
 - D. It is an irreversible and progressive change
- 80) Which of the following conditions is associated cell proliferation?
- A. Hypertrophy
 - B. Atrophy
 - C. Hyperplasia
 - D. Metastatic calcification
- 81) Programmed cell death is called
- A. Necrosis
 - B. Metaplasia
 - C. Amyloidosis
 - D. Apoptosis
- 82) Autoimmunity is a state in which the body display immunologic reactivity to itself.
- A. True
 - B. False
- 83) Pathological atrophy is seen in one of the following:
- A. Hypertensive heart disease
 - B. poliomyelitis
 - C. Bilharziasis
 - D. Sharp broken tooth
- 84) Which of the following processes elicits inflammation?
- A. Apoptosis
 - B. Necrosis
 - C. Hyperplasia
 - D. Metastatic calcification
- 85) Local signs and symptoms of acute inflammation include all the following EXCEPT:
- A. Redness
 - B. Hotness
 - C. Pain
 - D. Scar
- 86) Abscess is a type of
- A. Acute inflammation

- B. Chronic inflammation
- C. Diffuse inflammation
- D. Granuloma

87) The most important cell of acute inflammation is:

- A. Plasma cell
- B. Macrophage
- C. Neutrophil
- D. Lymphocyte

88) Factors affecting efficiency of repair includes:

- A. Age
- B. Nutritional status
- C. Extent of tissue damage
- D. All of the alternatives

89) A female patient went to the emergency room complaining of severe abdominal pain and admitted for surgical exploration. Which type of healing occurs in this surgical incision?

- A. Healing by primary intention
- B. Healing by secondary intention
- C. Healing by gliosis
- D. All of the alternatives

90) Complications of wound healing include all of the following **except**

- A. Keloid
- B. Ulcer
- C. Regeneration
- D. Cosmetic deformity

91) Each of the following is an example of hyperplasia except:

- A. Enlargement of one kidney after surgical removal of the other kidney
- B. Changes in the thyroid gland in response to increased demand for thyroid hormones
- C. Changes in breast tissue during pregnancy
- D. Enlargement of lymph nodes during a viral infection
- E. Enlargement of skeletal muscles after weight training

92) Intracellular system(s) sensitive to cell injury is/are:

- A. Aerobic respiration
- B. Maintenance of the integrity of cell membrane
- C. Synthesis of protein
- D. Integrity of genetic apparatus
- E. All of the alternatives

93) Which of the following can undergo apoptosis:

- A. Cells infected with virus
- B. Cells with DNA damage
- C. Increased oxidant within the cell

- D. None of the above
E. All of the alternatives
- 94) The following is true about hyperplasia **except**:
- A. Can be physiological
 - B. Is a precancerous condition
 - C. Is reversible
 - D. Is due to excess hormone stimulation
 - E. Can be associated with hypertrophy
- 95) Which of the following will show hypertrophy:
- A. The uterine myometrium in pregnancy
 - B. The female breast at puberty
 - C. The liver following partial resection
 - D. The ovary following menopause
 - E. The cervix with chronic inflammation
- 96) Hypertrophy is most closely associated with:
- A. Hypercalcemia
 - B. Necrosis
 - C. Chronic Irritation
 - D. Diminished Blood Supply
 - E. Increased Work Load
- 97) Hypertensive Enlargement of the heart is a form of:
- A. Fatty infiltration
 - B. Hypoplasia
 - C. Glycogen storage
 - D. Hypertrophy
 - E. Hyperplasia
- 98) Cigarette Smoking does not cause:
- A. Squamous Metaplasia
 - B. Smooth Muscle Hyperplasia
 - C. Defective Ciliary Action
 - D. Damage of Airway
 - E. Inhibition of Alveolar Leukocytes & Macrophages
- 99) Which of the following is (are) true about apoptosis:
- A. It is programmed cell death
 - B. It requires cell energy
 - C. Induces a severe inflammatory reaction
 - D. Both A and B
 - E. Both B and C
- 100) The following are known causes of Atrophy except:
- A. Aging
 - B. Starvation
 - C. Increased trophic hormone secretion
 - D. Immobilization
 - E. Denervation

- 101) A cardinal sign of Inflammation is:
- A. Vasoconstriction
 - B. Redness
 - C. Leukocyte Margination
 - D. Vasodilation
 - E. Slowing of the Circulation
- 102) Which one of the following is a harmful effect of acute inflammation:
- A. Dilution of toxins
 - B. Phagocytosis
 - C. Formation of fibrin
 - D. Swelling of tissue
 - E. Stimulation of immune response
- 103) Which of the following Cellular adaptation is seen after partial resection of liver:
- A. Hyperplasia
 - B. Hypertrophy
 - C. Metaplasia
 - D. Dysplasia
 - E. Atrophy
- 104) Which is not a major goal of the inflammatory response?
- A. increase blood flow to the site of injury
 - B. form an antigen-antibody response
 - C. dilute harmful substances at the site of injury
 - D. remove injured tissue
- 105) Inflammation may have beneficial effects; it works by diluting, destroying, or neutralizing harmful agents like microbes and toxins.
- A. True
 - B. False
- 106) In inflammation increased temperature (heat) is seen due to
- A. Decrease blood flow through the region
 - B. Increase blood flow through the region
 - C. Contraction of blood vessels
 - D. None of the alternatives
- 107) Which of the following events in acute inflammation occurs first?
- A. Redness
 - B. Heat
 - C. Swelling
 - D. Pain
 - E. Loss of function
- 108) Following are required for wound healing except
- A. Zinc
 - B. Copper
 - C. Vitamin C
 - D. Calcium

- E. None of the alternatives
- 109) The following stimuli can cause inflammation except:
- A. Hypoxia
 - B. Microbes
 - C. Tissue necrosis
 - D. Nutritional deficiency
 - E. Foreign bodies
- 110) Local signs of acute inflammation include:
- A. Redness
 - B. Pain
 - C. Swelling
 - D. Heat
 - E. All of the alternatives
- 111) Vascular change in response to inflammation includes all of the following except:
- A. Increased blood flow
 - B. Increased permeability
 - C. Decreased blood flow
 - D. Leakage proteins
 - E. None of the above
- 112) Predominant cells in chronic inflammation are:
- A. Neutrophils
 - B. RBCs
 - C. Macrophages
 - D. Basophils
 - E. None of alternatives
- 113) Metaplasia
- A. Is irreversible
 - B. Is commonly a change from squamous to columnar epithelium
 - C. An example is the transformation of epithelial cells into chondroblasts
 - D. Retinoids may play a role
 - E. Even if the stimuli is persistent it is a benign lesion
- 114) In apoptosis
- A. It involves physiologic and pathologic stimuli
 - B. Histologically, it involves coagulation necrosis
 - C. Its DNA breakdown is random and diffuse
 - D. Its mechanism involves ATP depletion
 - E. It involves an inflammatory tissue reaction
- 115) Which is incorrect in regards to apoptosis
- A. It may be regarded as a normal physiological process
 - B. It is characterized by chromatin condensation
 - C. It often elicits a strong inflammatory response
 - D. It is the process by which ovaries atrophy in post-menopausal women
 - E. It is characterized by cell shrinkage

- 116) Metaplasia is
- A. Reversible change from one cell type to another
 - B. Irreversible change from one cell type to another
 - C. Reduced function of cell
 - D. Increase in the number of cells
 - E. Increase in the size and function of cells
- 117) In necrosis
- A. The nuclear changes are due to non-specific breakdown of DNA
 - B. Karyolysis and pyknosis are the only 2 types of nuclear changes
 - C. In caseous necrosis the basic outline of the cells is preserved
 - D. There is a decreased eosinophilia in the necrotic cells
 - E. Liquefaction necrosis is characteristic of hypoxic injury
- 118) Infectious agents cause tissue damage through
- A. release enzymes that degrade tissue component
 - B. release toxins that kill cells
 - C. enter host cells and cause direct cell death
 - D. all of the above
- 119) In irreversible cell injury, there is
- A. loss of proteins and essential coenzymes
 - B. loss of ribonucleic acid
 - C. leakage of lysosomal enzymes
 - D. further depletion of intracellular ATPase
 - E. all of the above
- 120) Which of the following is not an aspect of the pathophysiology in disease.
- A. Causes
 - B. Pathogenesis
 - C. Morphological changes
 - D. Therapeutic treatment